

# Beatriz Dantas

Industrial AI ■ Hybrid Modeling ■ Scientific Research

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LinkedIn | GitHub | Google Scholar

## RESEARCH PROFILE

Industrial AI researcher with a Ph.D. in Chemical Engineering and a record of developing hybrid physics/AI methods and rigorously evaluated computational frameworks, supported by peer-reviewed publications. Integrates first-principles knowledge, industrial time-series data, simulation, and optimization to solve problems in energy and manufacturing.

## CORE EXPERTISE

|                                  |                                                                                                                                |
|----------------------------------|--------------------------------------------------------------------------------------------------------------------------------|
| <b>Industrial AI</b>             | Physics-informed ML, hybrid models, RNN/LSTM, transfer learning, surrogate modeling, industrial time series                    |
| <b>Research &amp; evaluation</b> | Algorithm development, computational experiments, benchmark comparison, statistical validation, and technical communication    |
| <b>Simulation</b>                | Dynamic process models, digital twins, mechanistic balances, optimization, operability, flexibility and safety-aware decisions |
| <b>Software &amp; cloud</b>      | Python (NumPy, pandas, SciPy, scikit-learn, PyTorch, TensorFlow, Pyomo), C#, MATLAB, SQL, Git, AWS                             |
| <b>Industrial platforms</b>      | AVEVA Process Simulation, AVEVA PI System/AF/SDK/Vision, Aspen Plus, Aspen IP.21, Ansys CFX                                    |

## PROFESSIONAL EXPERIENCE

### CVector Energy Corp.

New York, NY

*Research Engineer – Modeling & Analysis*

*Jul 2026 – Present*

- Develop and validate physics-based and data-driven models using real-time industrial data to support operational and economic decision-making.
- Analyze operating scenarios against equipment and facility constraints, quantifying performance, risk, and economic impact for margin-optimization recommendations.
- Translate custom engineering models into scalable workflows and ML surrogates with software, product, and domain teams.

### Dow

Plaquemine, LA

*ADISE Intern – Advanced Digital Internships for Scientists and Engineers*

*May 2025 – Aug 2025*

- Led data-driven troubleshooting of an extractive-distillation process using PCA/PLS, clustering, and industrial time-series analysis.
- Designed plant trials and visual analytics with process, automation, and data-science teams, supporting multimillion-dollar savings and a patent filing.

### Accenture

Campina Grande, Brazil

*Automation Analyst*

*Nov 2020 – Jun 2022*

- Developed industrial-data applications with AVEVA PI System, PI AF/SDK, and Aspen IP.21 for reliable acquisition, contextualization, and governance.
- Built automated analytics, monitoring tools, asset models, and dashboards for real-time manufacturing decisions.

## SELECTED RESEARCH CONTRIBUTIONS

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### Hybrid AI framework for nonlinear process flexibility

*Research lead | Python/Pyomo*

- Conceived a learning-and-optimization framework combining outer approximation, multiparametric LP/QP, neural-network reformulation, and adaptive sampling.
- Built benchmark datasets and comparisons across three case studies; achieved 99.17–99.36% accuracy and over 10× fewer errors on a dynamic electrolyzer than standard baselines.

### Physics-informed AI for dynamic hydrogen production

*Research lead | Python/PyTorch/TensorFlow*

- Created a recurrent architecture integrating LSTMs with mechanistic mass and energy balances for transient process prediction.
- Built the training and validation pipeline; reduced error by 92%, accelerated training by 5.4×, and cut new-data needs by 47%.

### Safety-aware process intelligence

*Simulation | optimization | uncertainty*

- Developed operability-based methods integrating process design, uncertainty, and risk; identified an operating region producing low-risk outcomes in 79% of evaluated scenarios.
- Extended scientific-computing methods to intensified chemical production, hazardous-release modeling, and membrane-based direct-air capture.

## PATENT APPLICATIONS

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- **Lead inventor**, “Prediction of Overhead Composition in an Extractive Distillation Column.” U.S. application filed; Dow TDCC No. 87276-US-PSP.
- “Real-Time Safety-Aware Control and Monitoring System Design.” U.S. Provisional Patent Application No. 63/951,033.

## EDUCATION & RESEARCH LEADERSHIP

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### West Virginia University

**Morgantown, WV**

*Ph.D., Chemical Engineering; GPA: 4.0/4.0*

*May 2026*

- Dissertation: *Hybrid Mechanistic/Data-Driven Modeling, Design, and Optimization of Safe, Sustainable, and Flexible Process Systems*.
- Mentored RNN and process-control research; mentee earned 1st Place in AIChE CAST and 2nd Place Overall at the 2024 AIChE Annual Meeting.
- Peer reviewer for *Digital Chemical Engineering* and *Industrial & Engineering Chemistry Research*.

### Federal University of Campina Grande

**Campina Grande, Brazil**

*B.Sc., Chemical Engineering*

*Apr 2022*

## SELECTED PUBLICATIONS & RECOGNITION

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- Dantas et al., “A hybrid framework for flexibility analysis of nonlinear process systems,” *Computers & Chemical Engineering*, 2026.
- Dantas et al., “Hybrid mechanistic/data-driven modeling for enhancing dynamic operations in electrolysis-based hydrogen production,” *International Journal of Hydrogen Energy*, 2026.
- Additional peer-reviewed work in *AIChE Journal*, *ACS Engineering Au*, *Nature Chemical Engineering*, and *Methods in Chemical Process Safety*.
- Statler College Excellence in Research – Ph.D.; Outstanding Chemical Engineering Graduate Student (2026); AIChE CAST Best Poster Semi-Finalist (2025).

**Languages:** English (fluent), Portuguese (native)

**Work arrangement:** Available for hybrid work